

MSSC (Microbial Sump-to-Soil Cultivator)

Prior Art & IP Defense Dossier

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Date: August 27, 2025 (America/Chicago)

1. Executive Summary

- Earliest infrastructure groundwork: duck pond + bog filter + multi-species manures discussed March 22, 2025.
- First explicit MSSC framing: March 23–24, 2025 timeframe.
- Compost tea & microbial cycling detailed through late March, including ducks as biological primers (March 28, 2025).
- Patent-style draft authored June 8–10, 2025; experimental logs initiated June 10, 2025.

2. Chronological Timeline & Evidentiary Index

Date	Record / Evidence Summary
2025-03-22	Duck pond + bog filter under construction; access to multi-species manures (duck, chicken, go
2025-03-23	MSSC framing begins (sump–soil cycling, compost tea intention).
2025-03-24	MSSC explicitly acknowledged as coming into fruition.
2025-03-28	Ducks as biological primers for rows—rotation-of-life concept.
2025-06-08	Rewrite MSSC wiki w/ validated components; align with Tri-Source.
2025-06-09	HomeNode concept: residential nitrate removal using MSSC biome.
2025-06-10	Lab tests: small containers; begin documentation & scale plan.

3. Patent-Style Specification (June 2025)

Field

Agricultural microbiology; integrated water treatment and soil amendment systems.

Summary

MSSC bioreacts runoff via sump/bog reactor and returns inoculated irrigation; captures compost tea stream.

Representative Embodiment

60-ft pond, 10×40-ft bog, aeration, dosing ports, recirc pump, drip tee, compost tea bleed.

Controls

DO/ORP/EC/pH/temp sensors; stage-specific logic.

Duck-Row Priming

Rotational duck priming to establish beneficial consortia.